Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
END Semester Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours
Max Marks : 40

Course Code: UCS619
Date: May 31, 2025
Name of Faculty: Dr. Ajay Kumar, Dr. Vaibhav Pandey, Dr. Samya Muhuri

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Note: Attempt any **five** questions. Assume missing data, if any, suitably. Give appropriate reasoning for your answers.

1. (a) Consider a Y error occurs at 9th qubit. Explain the status of the quantum system at each step of Shor's 9-qubit code. **(6, CO3, BL3)**

   (b) Write down the matrix for Quantum Fourier Transform (QFT) for 2 qubits (QFT_2). Write down the matrix form of $H^{\otimes 2}$. Now check whether QFT_2 is equal to $H^{\otimes 2}$ or not. **(2, CO3, BL2)**

2. Given the search space $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ (that is, we are given a function $f : \sum^3 \rightarrow \sum$ where $\sum = \{0, 1\}$). Suppose $|111\rangle$ and $|011\rangle$ are marked as present (in other words, $f(111) = f(011) = 1$). Calculate the different steps of Grover's algorithm to find the element $|111\rangle$ and $|011\rangle$ in the search space. **(8, CO4, BL3)**

3. (a) Discuss the advantages of B92 protocol over BB84 protocol. **(2, CO4, BL1)**

   (b) In B92 protocol, state the four rules based on which Bob decides his observation in + basis or × basis. **(2, CO4, BL1)**

   (c) Based on the B92 protocol and given information, illustrate the Bob's observations and Bob's bits in the following table. In case of two outputs, mention both (for example, in case 0 and 1, mention both 0|1) mention both outputs separated by | . **(4, CO4, BL4)**

Bit number	1	2	3	4	5	6	7	8
Alice's random bits	↗	→	↗	→	↗	↗	↗	→
Quantum channel	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓
Bob's random bases	+	X	+	+	X	+	X	+
Bob's observations								
Bob's bits								

4. (a) Deutsch-Jozsa algorithm is used to check whether a function is constant or balanced. Draw the circuit of Deutsch-Jozsa's algorithm and explain the value at each point for the following scenario of three qubits: $f(000) = f(001) = f(100) = f(101) = 1$, $f(010) = f(011) = f(110) = f(111) = 0$. **(6, CO4, BL3)**

P.T.O.

(b) Examine the output when S-gate is operated on $|+\rangle$ and $|-\rangle$. **(2, CO2, BL3)**

5. (a) Explain the procedure of superdense coding, assuming that the third party generates the entangled pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. **(5, CO1, BL3)**

(b) The standard way to write the state of a qubit using the bloch sphere is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

Explore the value of θ and ϕ for

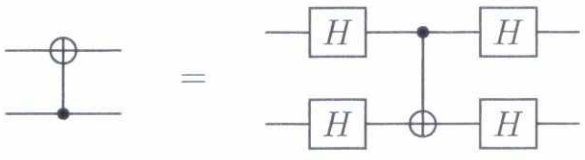
A) $|+\rangle$ state

B) $|i\rangle$ state

C) $|\psi\rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix}$

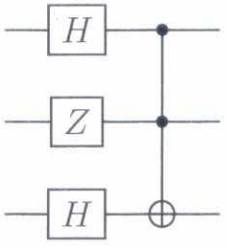
(3, CO1, BL3)

6. (a) Prove that the following two circuits are equivalent to each other. **(3, CO2, BL3)**



(b) Consider a new gate $\boldsymbol{W}$, such that $\boldsymbol{W} = \frac{(1-i)}{2}(\boldsymbol{I} + i\boldsymbol{T})$. Write the matrix equivalent of $\boldsymbol{W}$ gate. **(3, CO2, BL3)**

(c) Show the matrix equivalent for the given figure.



(2, CO2, BL3)